

DETROIT WAYNE COUNTY AP, MI, USA

WMO: 725370

Lat: 42.231N

Lon: 83.331W

Elev: 631

StdP: 14.36

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94847

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.1	7.3	-6.8	3.9	2.9	-1.6	5.2	9.0	28.1	27.7	25.4	29.7	9.2	250	0.534

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.1	90.3	74.0	87.5	72.4	85.0	70.9	76.7	86.2	75.0	83.5	73.3	81.5	12.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.7	128.5	81.9	72.2	121.8	80.1	70.5	115.0	78.5	40.4	86.4	38.8	83.6	37.3	81.7	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
24.8	21.0	18.8	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-3.6	95.0	6.9	2.7	-8.5	96.9	-12.5	98.5	-16.4	100.0	-21.3	101.9		
			-4.4	79.8	6.7	2.0	-9.2	81.3	-13.1	82.5	-16.9	83.6	-21.7	85.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.8	25.5	28.1	37.2	49.1	60.3	70.1	74.1	72.5	65.5	53.4	41.4	31.3
	DBStd	19.09	11.04	10.22	10.47	8.88	8.35	6.62	5.41	5.04	7.19	8.35	8.88	9.35
	HDD50	2848	760	615	418	120	10	0	0	0	1	58	284	582
	HDD65	6036	1224	1034	862	482	197	26	2	5	80	372	708	1044
	CDD50	3152	1	2	22	92	329	604	746	698	467	162	26	3
	CDD65	868	0	0	1	5	51	181	283	238	97	12	0	0
	CDH74	6967	0	0	10	67	466	1494	2444	1736	668	81	1	0
	CDH80	2136	0	0	1	10	122	498	829	485	178	13	0	0
Wind	WSAvg	8.9	10.6	10.1	9.8	10.0	8.7	7.8	7.4	6.9	7.2	8.7	9.7	9.8
Precipitation	PrecAvg	32.6	2.2	2.0	2.2	3.0	3.5	3.3	3.2	3.2	3.0	2.6	2.3	2.1
	PrecMax	43.1	4.1	4.6	4.7	5.3	7.8	5.8	5.9	5.9	5.8	6.9	5.7	4.6
	PrecMin	24.1	0.6	0.7	0.6	0.6	1.1	1.1	1.4	0.3	0.8	0.2	0.6	0.6
	PrecStd	4.4	0.9	0.9	1.1	1.1	1.7	1.3	1.2	1.5	1.6	1.3	1.1	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.6	61.2	73.9	81.4	87.5	93.2	94.5	92.1	89.9	81.5	68.3	59.6
		MCWB	53.4	53.5	61.0	64.9	70.0	74.5	76.8	75.1	73.1	68.0	57.0	55.5
	2%	DB	51.0	52.5	65.4	74.8	83.5	89.1	90.6	87.9	85.1	75.8	62.8	53.6
		MCWB	48.4	47.2	54.8	59.8	67.7	72.6	74.5	73.4	70.5	64.6	55.8	50.0
	5%	DB	45.1	46.1	58.8	69.6	79.7	85.7	87.5	85.0	80.9	71.3	59.2	48.3
		MCWB	42.0	41.6	51.4	56.5	66.3	70.8	72.7	71.9	68.4	61.8	53.4	44.4
	10%	DB	39.7	41.1	53.1	64.4	75.0	82.3	84.7	82.5	77.5	67.2	54.9	43.4
		MCWB	37.1	37.1	46.3	53.7	63.5	69.0	71.1	70.3	66.7	59.2	49.5	40.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.0	55.3	62.2	66.4	73.1	77.4	79.8	78.4	75.8	70.6	61.0	56.1
		MCDB	56.3	58.5	72.8	78.3	84.1	89.7	90.3	88.5	85.6	79.1	64.7	59.0
	2%	WB	49.0	48.0	56.8	62.4	70.3	74.9	77.1	76.1	73.2	67.0	57.4	51.0
		MCDB	50.8	52.1	62.8	71.2	79.7	85.4	86.8	84.2	81.1	72.9	61.4	53.2
	5%	WB	42.1	41.6	52.4	58.5	68.0	73.1	75.1	74.3	70.9	63.5	53.7	45.3
		MCDB	44.6	45.6	58.4	67.2	76.8	82.8	83.5	81.4	77.6	69.6	58.1	48.5
	10%	WB	37.3	37.4	46.5	54.9	65.2	71.2	73.3	72.6	68.8	60.2	50.0	40.3
		MCDB	39.4	40.7	52.3	63.1	73.5	79.4	81.4	79.4	75.1	65.9	54.5	43.2
Mean Daily Temperature Range	5% DB	MDBR	12.6	13.6	16.3	18.7	18.8	18.2	18.1	17.0	18.0	16.7	13.9	11.2
		MCDBR	16.9	20.2	24.8	26.4	23.1	21.4	20.2	19.1	21.4	22.0	19.5	16.1
		MCWBR	15.3	16.3	16.7	15.8	11.9	9.8	9.2	8.8	10.0	12.5	14.4	13.8
	5% WB	MCDBR	16.5	19.3	23.4	23.7	20.6	18.8	18.1	16.7	18.5	19.2	17.9	15.4
		MCWBR	15.7	16.3	17.3	16.2	12.2	9.8	9.5	9.0	9.9	12.5	14.7	14.3
Clear-Sky Solar Irradiance	taub	0.302	0.338	0.355	0.402	0.436	0.452	0.443	0.436	0.392	0.361	0.319	0.313	
	taud	2.395	2.273	2.342	2.232	2.188	2.183	2.233	2.242	2.343	2.410	2.493	2.400	
	Ebn at Noon	266	271	282	275	267	262	264	261	266	261	259	250	
	Edh at Noon	25	33	35	43	46	46	43	42	35	29	23	23	
All-Sky Solar Radiation	RadAvg	482	760	1114	1463	1708	1922	1931	1674	1355	883	553	387	
	RadStd	46	94	110	127	148	123	98	64	118	75	72	33	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+151	N/A
(47)	Regional (64 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+148	N/A

Nomenclature: See separate page